

Additional practicals

The practicals listed below are mentioned in the specification as “further suggested practical”. Just skim through them and re-read the specification to confirm if all suggested practicals are covered here since I may have missed some.

Additional Practical 1: Thermal Decomposition of Group 1 and Group 2 Nitrates/Carbonates

Purpose:

To compare the thermal stability of nitrates and carbonates of Group 1 and Group 2 elements.

Method Overview:

- Heat small spatula quantities of different metal nitrates or carbonates in test tubes.
- Observe decomposition products:
 - **Group 1 nitrates (e.g., KNO_3):** Decompose to **metal nitrite (MNO_2)** and **oxygen**.
 - **Group 2 nitrates (e.g., $\text{Mg}(\text{NO}_3)_2$):** Decompose to **metal oxide, nitrogen dioxide**, and **oxygen**.
 - **Carbonates:** Generally decompose to **metal oxides + CO_2** (Group 2 only).

Tests:

- **Oxygen:** Relights glowing splint.
- **CO₂:** Turns limewater milky.
- **NO₂:** Brown gas with sharp smell (toxic).

Key Concept:

Thermal stability increases **down the group** — heavier cations have a **lower charge density** and polarise the anion less.

Additional Practical 2: Determining Solubility of Calcium Hydroxide by Titration

Purpose:

To determine the solubility (mol dm^{-3}) of sparingly soluble Ca(OH)_2 in water.

Method Overview:

- Prepare saturated solution of Ca(OH)_2 by stirring solid in water and filtering off undissolved excess.
- Pipette 25.0 cm^3 of this solution into a conical flask.
- Titrate against standardised HCl using methyl orange.
- Repeat for concordant results.

Calculation:

Use moles of HCl to find moles of OH^- (2 per Ca^{2+}), then calculate solubility of Ca(OH)_2 .

Key Concept:

This uses a **back titration** to determine the concentration of a substance that cannot be directly weighed out in solution.

Additional Practical 3: Determination of Water of Crystallisation by Titration

Purpose:

To determine the number of water molecules (x) in a salt like $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$.

Method Overview:

- Weigh a known mass of hydrated sodium carbonate.

- Dissolve in a known volume of water to make a solution.
- Titrate 25.0 cm³ aliquots against standardised HCl using methyl orange.
- Calculate moles of Na₂CO₃ reacting using stoichiometry.

Calculation:

- Moles of Na₂CO₃ from titration.
- Mass used = total from balance.
- Compare actual molar mass with calculated one to find **x**.

Key Concept:

Titration data allows you to work backwards to determine the molecular formula.

Additional Practical 4: Reaction of Solid Halides with Concentrated Sulfuric Acid

Purpose:

To distinguish chloride, bromide, and iodide salts based on redox behaviour and gas evolution when reacted with conc. H₂SO₄.

Method Overview:

- Add conc. H₂SO₄ dropwise to small amounts of KCl, KBr, and KI in fume cupboard.
- Observe:
 - **KCl:** Steamy HCl fumes (not redox, acid-base only).
 - **KBr:** Steamy HBr + brown Br₂ fumes (HBr is further oxidised).
 - **KI:** Steamy HI + purple I₂ vapour + red-brown solid + toxic **H₂S/S** (strongest redox).

Key Concept:

- **Reducing power of halide ions increases** down the group.
 - Only iodide can reduce H₂SO₄ to **H₂S**; bromide to **SO₂**.
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Additional Practical 5: Enthalpy of Combustion of an Alcohol

Purpose:

To determine the enthalpy change of combustion for alcohols like ethanol or propan-1-ol.

Method Overview:

- Measure mass of spirit burner before and after combustion.

- Burn under a **calorimeter containing water**.
- Record temperature rise in a known mass of water.
- Repeat with other alcohols for comparison.

Calculation:

- Use $q = mc\Delta T$ for water.
- Find moles of alcohol burned and convert to kJ mol^{-1} .

Sources of Error:

- Heat lost to surroundings.
- Incomplete combustion (soot/CO formation).
- Use of draught shield improves accuracy.

Additional Practical 6: Enthalpy of Hydration of Anhydrous Copper(II) Sulfate

Purpose:

To determine the enthalpy change of hydration of Cu^{2+} ions indirectly using Hess's Law.

Method Overview:

- Measure $\Delta H(\text{solution})$ for:
 - Anhydrous CuSO_4
 - Hydrated $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- Add known masses of each to water and record temperature changes.

Calculation:

- Use $q = mc\Delta T$ for both.
- Apply Hess's cycle: $\Delta H_3 = \Delta H_1 - \Delta H_2$
where ΔH_3 = enthalpy of hydration.

Key Concept:

The enthalpy of hydration cannot be measured directly, so **indirect method with Hess's Law** is used.

Additional Practical 7: Determination of Enthalpy Change for CaCO_3 Decomposition

Purpose:

To determine the enthalpy change of decomposition of CaCO_3 using Hess's Law.

Method Overview:

- Measure ΔH for:
 - $\text{CaCO}_3 + \text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$
 - $\text{CaO} + \text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}$
- Use a polystyrene cup and thermometer.
- Record temperature changes after adding solid to acid.

Calculation:

- $q = mc\Delta T$ for both reactions.
- Apply Hess's Law to determine: $\Delta H (\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2) = \Delta H_1 - \Delta H_2$

Key Concept:

Again, decomposition cannot be measured directly, so **two experimental enthalpy changes** are used instead.

Additional Practical 8: Investigating Rate of Reaction (Surface Area, Temperature, Concentration)

Purpose:

To investigate how changing physical factors affects reaction rate.

Examples:

1. **Marble chips (CaCO_3) + HCl:**
 - Vary **surface area** (powder vs lumps).
 - Measure gas volume with syringe or displacement.
2. **Sodium thiosulfate + HCl:**
 - Vary **concentration** or **temperature**.
 - Record time until **cross disappears** (precipitate forms).
3. **Hydrogen peroxide decomposition:**
 - Vary **catalyst mass** (MnO_2).
 - Measure **O₂ gas** collected over time.

Key Concept:

Rate $\propto$ frequency of successful collisions; supports **collision theory** and basic rate law interpretations.

Additional Practical 9: Investigating Equilibrium Position (Le Chatelier's Principle)

Purpose:

To observe how changes in concentration or temperature affect the position of equilibrium.

Example System:

- NO_2 (brown gas) $\rightleftharpoons$ N_2O_4 (colourless)

Method Overview:

- Place equilibrium mixture in a sealed gas syringe or boiling tube.
- Vary **temperature**:
 - Cooling shifts equilibrium toward N_2O_4 (pale).
 - Heating shifts equilibrium toward NO_2 (dark brown).
- Vary **concentration** (in other systems like chromate $\rightleftharpoons$ dichromate):
 - Add acid or alkali to shift colour equilibrium.

Key Concept:

Le Chatelier's Principle.

Additional Practical 10: Preparation of 1-Bromobutane**Purpose:**

To prepare a primary halogenoalkane via **nucleophilic substitution** of an alcohol.

Reagents:

- Butan-1-ol
- Potassium bromide
- Concentrated sulfuric acid

Method Overview:

- Reflux mixture of butan-1-ol, KBr, and H_2SO_4 .
- Br^- is protonated to form HBr **in situ**, which reacts with the alcohol.
- Distill to isolate 1-bromobutane (b.p. $\sim 102^\circ\text{C}$).
- Use a **separating funnel** to purify product and remove aqueous layer.
- Dry with anhydrous CaCl_2 , then redistill.

Key Concept:

Primary alcohol + HBr $\rightarrow$ primary bromoalkane

Additional Practical 11: Preparation of Cyclohexene from Cyclohexanol

Purpose:

To prepare an alkene by **acid-catalysed dehydration** of an alcohol.

Reagents:

- Cyclohexanol
- Concentrated phosphoric(V) acid

Method Overview:

- Mix cyclohexanol and H_3PO_4 in a **round-bottom flask**.
- Add **anti-bumping granules**, heat under reflux.
- Collect **cyclohexene by distillation** (b.p. $\sim 83^\circ\text{C}$).
- Use separating funnel to wash with water, NaHCO_3 (neutralise acid), then water again.
- Dry and redistill.

Test for Product:

- Bromine water decolorised — confirms $\text{C}=\text{C}$ bond.

Key Concept:

E1 mechanism (or E2), elimination of water from secondary alcohol.

Additional Practical 12: Preparation of Limonene from Orange Peel (Steam Distillation)

Purpose:

To extract limonene (a terpene) from citrus peel using **steam distillation** — a “green chemistry” example.

Method Overview:

- Chop and crush orange peel.
- Place in distillation apparatus with **water**.
- Heat gently to allow **steam to carry volatile oils** over.
- Collect **oil-water mixture** in a receiver.
- Use **solvent extraction** to isolate limonene (e.g., with hexane).
- Dry with anhydrous salt and evaporate solvent.

Key Concept:

Steam distillation allows separation of **heat-sensitive, water-insoluble** organic compounds without decomposition.

Additional Practical 13: Reactions of Alcohols with Halide Reagents

Purpose:

To prepare halogenoalkanes from alcohols using different halogenation reagents.

Method Overview:

(a) With PCl_5 :

- Add PCl_5 to a dry alcohol in a fume cupboard.
- Observe **steamy HCl fumes**.
- Test with damp blue litmus – turns red then bleached.
- Confirms presence of **–OH group**.

(b) With $\text{KBr} + \text{H}_2\text{SO}_4$:

- Mix alcohol, **KBr**, and **conc. H_2SO_4** (which generates HBr in situ).
- Heat under reflux.
- Bromination via nucleophilic substitution.

(c) With Red Phosphorus and Iodine:

- Heat alcohol with red P and I_2 .
- Forms **phosphorus(III) iodide (PI_3)** in situ, which reacts to form iodoalkane.

Key Concept:

These reactions demonstrate different synthetic routes for halogenoalkane formation and provide qualitative evidence for the hydroxyl group.

Additional Practical 14: Comparing Reactions of Primary and Secondary Alcohols

Purpose:

To distinguish primary and secondary alcohols using **oxidation reactions**.

Method Overview:

- Add **acidified potassium dichromate(VI)** ($\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4$) and warm:
 - **Primary alcohol:** Orange → green; can form **aldehyde**, further to carboxylic acid.

- **Secondary alcohol:** Orange → green; forms **ketone**.
- **Tertiary alcohol:** No reaction; remains orange.

Further Tests:

- Use **Tollens' reagent**:
 - Aldehyde: silver mirror.
 - Ketone: no reaction.
- Use **Fehling's solution**:
 - Aldehyde: brick red precipitate.
 - Ketone: no reaction.

Key Concept:

Primary vs secondary alcohols differ in oxidation behaviour — this helps confirm their identity.

Additional Practical 15: Determination of a Formula or Equation by Experiment

Purpose:

To find an empirical formula or confirm a balanced chemical equation via experimental data.

Method Overview:

- **Empirical formula:** E.g., heat a known mass of a metal oxide (like CuO) in hydrogen or methane atmosphere.
 - Measure mass before and after reduction to copper.
 - Use mass loss (oxygen) to calculate the formula.
- **Confirming equation:** Use mass or gas volume changes to verify mole ratios in reactions like:
 - $\text{Mg} + \text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$

Key Concept:

Experimental stoichiometry supports theoretical formulas and balancing of chemical equations.

Additional Practical 16: Preparation of a Salt and Determining Percentage Yield

Purpose:

To synthesise a salt and evaluate efficiency via **% yield**.

Example:

- Prepare ammonium iron(II) sulfate from $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ and H_2SO_4 .

Method Overview:

- Dissolve measured mass of iron(II) sulfate in **warm dilute sulfuric acid**.
- Cool to crystallise.
- Filter, dry crystals.
- Weigh and calculate **percentage yield**:
 - $\% \text{ yield} = (\text{actual mass} / \text{theoretical mass}) \times 100$

Key Concept:

purification (crystallisation) and quantitative analysis.

Additional Practical 17: Cracking of Hydrocarbons (e.g., Liquid Paraffin)

Purpose:

To demonstrate **thermal cracking** and formation of alkenes.

Method Overview:

- Heat **liquid paraffin** soaked on mineral wool in boiling tube.
- Use **aluminium oxide** (Al_2O_3) as a catalyst.
- Collect gaseous products over water in test tubes.
- Test gases:
 - **Bromine water**: decolorised → confirms alkene.

Key Concept:

Shows conversion of large alkanes to smaller alkanes and alkenes via **thermal decomposition** using a solid catalyst.

Additional Practical 18: Solubility of Simple Molecules in Different Solvents

Purpose:

To explore how intermolecular forces influence solubility of substances in different types of solvents.

Method Overview:

- Choose representative molecules (e.g., **ethanol, hexane, halogenoalkanes**).
- Test solubility in:
 - **Water** (polar, hydrogen bonding),

- **Hexane** (non-polar),
- **Propanone** (polar aprotic).
- Observe which substances dissolve fully, partially, or not at all.

Key Concepts:

- **“Like dissolves like”**: polar solvents dissolve polar molecules (due to hydrogen bonding or dipole interactions); non-polar solvents dissolve non-polar molecules (due to London forces).
 - E.g., alcohols are miscible with water, halogenoalkanes are not.
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Additional Practical 19: Measuring Enthalpy Change of Vaporisation

Purpose:

To determine the energy required to vaporise a liquid such as water or ethanol.

Method Overview:

- Heat a known mass of liquid (e.g., water) using an electric heater or Bunsen burner.
- Measure:
 - Initial and final temperatures,
 - Energy supplied (from electrical energy or using $q = mc\Delta T$ if indirect),
 - Mass of liquid evaporated.

Calculation:

- Use:

$$\Delta H = \frac{q}{n}$$

where q = energy transferred, n = moles vaporised.

Sources of Error:

- Heat loss to surroundings,
 - Incomplete vaporisation,
 - Evaporation during weighing.
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Additional Practical 20: Measuring Temperature Change on Dissolving Substances

Purpose:

To determine whether dissolution is **endothermic or exothermic**, and measure the temperature change.

Method Overview:

- Weigh a known mass of solid (e.g., **NaCl**, **NH₄NO₃**, or **KCl**).
- Add to a known volume of water in a polystyrene cup.
- Stir and record temperature change.
- Repeat for different substances.

Calculation:

- Use: $Q = mc\Delta T$ and $\Delta H = \frac{q}{n}$

where q = energy transferred, n = moles vaporised

Key Concept:

- Dissolution can absorb or release heat depending on lattice and hydration enthalpies.